# CS 475/675 Machine Learning: Project

This semester you will complete an independent project to learn how machine learning can be applied in practice. You will work in groups of 3-4 students, and all students in the group will receive the same grade. The final project will be worth 10% (100 points) of your final grade.

You have a choice between two types of projects.

- **Application**: These projects will apply machine learning methods to a dataset to achieve the desired goals in an application domain. Here the desired goals may be a good predictive performance, but may also include other properties like robustness, fairness, safety, in addition to the predictive performance.
- **Methods**: These projects will conduct an analysis on certain properties of machine learning algorithms and develop novel techniques to improve the properties of state-of-the-art methods.

Additionally, each project will select a specific area: either an application domain or a subfield of ML methods. Each team will be assigned to a CA/TA/instructor who will provide feedback and review project deliverables.

## Project Structures

Your first goal will be to form a team of 3-4 students. You will share your grade with your team members, so **pick people with whom you can work well.**

The final project has several milestones and deliverables.

<u>0) Tell me who you will be working with: Due **3/30 Monday (11:59pm)**</u>

<u>1) Project proposal (20 points): Due **Wednesday 4/8 (11:59pm)**</u>
Your project proposal should be about 1 page. We will provide a template.

<u>2) Project presentation (60 points): **In the week of April 27th.**</u>
Each team will present a ~5 minute presentation on their project, followed by a 5 minute QA with teaching staff (instructors/TAs/CAs). Include an update on your deliverables: are you on track to accomplish everything you proposed? What have you learned in the project?

4) <u>Project report (Project git repo and Jupyter Notebook) (20 points): **Due 5/8 (11:59pm)**</u>
The final deliverable for your project will be a git repository containing all of your code and a Jupyter Notebook containing the final writeup. The git repo can be stored on Github.com and you will provide a link to the repository. You will also provide a link to a Jupyter notebook. The notebook can be stored directly in the git repo, or hosted on Google Colab. The notebook should be structured as a writeup, with text explanations mixed with a step by step walkthrough of the project, including summaries of the methods, goals, figures, and results. Here is a

template:
https://colab.research.google.com/drive/1X9kDSVd1S3XQliWImU8kiveyclaXw5V1?usp=sharing

Eventually, you will print a pdf and submit.

**We will provide a list of suggested ideas/data later, but you are encouraged to come up with your own ideas.**

## How are you going to be evaluated?

Projects will be evaluated primarily on how well students **apply concepts and methods learned in the course**, rather than solely on the final outcome or performance of the system. Given that you will have access to potential previous codes (like if you start with a research paper) or even AI, what we will focus on is your efforts **beyond** what is already available to you. Strong projects should demonstrate a clear connection between course material and the design of the project, including appropriate use of models, assumptions, evaluation methods, and theoretical principles discussed in class. Careful reasoning, sound experimental design, and thoughtful analysis of results are valued more than achieving the best performance. Our rubrics will cover things like whether the project shows clear evidence of conceptual understanding learned in the class, principled methodology, critical reflection on lessons learned in the process, and novelty in terms of the application or the methods used.